

JATIN GARG

B.Tech CSE Student | Full-Stack & Agentic-AI Developer

+91 6239454057 | jatin26garg@gmail.com | linkedin.com/in/jatin-garg-8b464a379 | github.com/jatin26garg
leetcode.com/u/jatin6garg | x.com/JatinGarg1090

SUMMARY

B.Tech (CSE) undergraduate at Thapar Institute of Technology with hands-on experience building full-stack web applications and Agentic-AI / RAG systems. Skilled in React, Next.js, Node.js and Python, with practical exposure to LLM frameworks, vector databases, and real-time systems (WebSockets/WebRTC). Strong problem-solving foundation with 500+ problems solved on LeetCode. Actively seeking an internship to apply and grow full-stack and applied-AI engineering skills.

EDUCATION

Thapar Institute of Engineering and Technology

B.Tech, Computer Science & Engineering — 2nd Year

India

CGPA: 8.79

- **JEE Mains:** Scored 98 percentile

PROJECTS

Sovereign RAG System (PDF-Rag) | Python, FastAPI, Qdrant, LangChain, Ollama (Qwen3-8B), BGE-M3

github.com/jatin26garg/PDF-Rag

- Built a self-hosted, air-gapped Retrieval-Augmented Generation (RAG) system that answers questions over PDF/DOCX/TXT documents with zero external API calls, ensuring full data sovereignty.
- Implemented hybrid retrieval combining dense and sparse BGE-M3 embeddings with Reciprocal Rank Fusion (RRF) for higher-quality, citation-backed answers, served via a FastAPI backend.
- Integrated a locally-hosted Qwen3-8B LLM through Ollama and Qdrant as the vector store, along with document ingestion (chunking + embedding) and a REST API for upload, query, and document management.
- Added automated tests, including verification that no data leaves the network, and documented setup/deployment for reproducibility.

Collaborative Code Editor | Next.js, TypeScript, Socket.IO (WebSockets), MongoDB, Tailwind CSS

github.com/jatin26garg/collaborativeCodeEditor

- Developed a real-time, multi-user collaborative code editor using Next.js and TypeScript, allowing multiple users to write and edit code simultaneously in shared sessions.
- Built a dedicated Socket.IO server to broadcast live code changes and cursor/room state between connected clients with low latency.
- Designed data models and persistence layer (Models directory) to store session/user data, and structured the UI with reusable React components and Tailwind CSS for a responsive editor experience.

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, TypeScript
- **Web / Frontend:** React, Next.js, HTML5, Tailwind CSS, Redux, Context API
- **Backend:** Node.js, Express.js, FastAPI, RESTful APIs, JWT & OAuth, WebSockets, WebRTC
- **Databases:** SQL, MongoDB, ChromaDB, Qdrant
- **Agentic-AI / RAG:** LangChain, LLMs (OpenAI, Claude, Hugging Face, Ollama Qwen3-8B), vector embeddings & chunking, PyPDF2, pdfplumber
- **Tools:** Git & GitHub, VS Code, Docker

PROBLEM SOLVING

- **LeetCode ([jatin6garg](https://leetcode.com/u/jatin6garg)):** 501+ problems solved, primarily in C++ — strong in Arrays, Dynamic Programming, Binary Search, Backtracking, and Hash Tables; earned 50-Day and 100-Day consistency badges.